JACOBUS PHILIP HAUPT

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SUMMARY

Computational scientist (PhD, Max Planck Institute) with fives years of research in Monte Carlo methods, stochastic optimisation and large-scale numerical computing, currently working on machine learning foundation models. I develop research code from theory to production in Python, Fortran and C++ on HPC clusters. Seeking quantitative researcher and developer roles applying statistical modelling and simulation to financial markets.

RESEARCH EXPERIENCE

Postdoctoral Researcher January 2026 – Present
Sorbonne University *Paris, France*

- Combining selected configuration interaction (an iterative sparse eigensolver) with diffusion Monte Carlo to generate high-accuracy training data for machine-learning foundation models of molecules for drug discovery
- Building ML training and evaluation pipelines in JAX and PyTorch; HPC development in Python and C++; developing workflows on the Aurora supercomputer
- Industrial and academic collaborations with Qubit Pharmaceuticals and Argonne National Lab

Doctoral Researcher, then Postdoctoral Researcher November 2020 – November 2025
Max Planck Institute for Solid State Research *Stuttgart, Germany*

- Developed stochastic algorithms for high-dimensional eigenvalue problems (from *ab initio* quantum chemistry), from mathematical derivation through to parallel implementation and publication
- Lead developer of PyTCHInt, a Fortran numerical-integration library with a Cython/Python interface; introduced CI/CD, unit testing and generated documentation
- Co-lead developer of NECI, the largest MPI-parallel Fortran code for full configuration interaction quantum Monte Carlo, a stochastic sparse-matrix eigensolver; introduced various improvements for transcorrelated methods, including a variance reduction method
- Designed and optimised many-parameter models under Monte Carlo noise using variational and stochastic optimisation (first-author, *J. Chem. Phys.* 2023)
- Statistical analysis of correlated Monte Carlo time series: error estimation via blocking analysis, bias and variance control, convergence diagnostics

SELECTED PERSONAL PROJECTS

CIPSIpy JAX, LLMs

- Implemented a selected configuration interaction solver from scratch in JAX: iterative subspace selection with perturbative importance ranking and sparse diagonalisation
- Benchmarked against reference results and fully unit-tested; methods and results documented online

Biotech Event Study Python, pandas, yfinance

- Proof-of-concept event study of biotech equity returns around FDA approval decisions using cumulative abnormal returns
- Found larger price reactions to unanticipated approvals than to expected ones

EDUCATION

PhD in Computational Chemistry 2020 – 2025
Max Planck Institute for Solid State Research / University of Stuttgart *Stuttgart, Germany*

- Thesis awarded *magna cum laude*: *Development of the Transcorrelated Full Configuration Interaction Quantum Monte Carlo Method*; advisor Prof. Ali Alavi

Doctoral Studies in Theoretical Physics 2019 – 2020
University of Toronto *Toronto, Canada*

- Direct-entry PhD scholarship; graduate coursework and research in computational electromagnetics

BSc in Combined Honours Physics & Mathematics, Minor in Computer Science
With Co-Operative Education 2014 – 2019
The University of British Columbia *Vancouver, Canada*

- GPA 85.8/100 (4.00), graduated with distinction; graduate-level coursework in mathematics and physics; advanced undergraduate coursework in computer science
- UBC Science Scholar, Palliser-Wilson Scholarship in Mathematics, Dean's Honour List (all years)

EARLIER EXPERIENCE

- **Research Intern, ML in atomic physics** (UBC, 2019): sparse regression (lasso) for model selection in quantum scattering; NSERC USRA award
- **Research Intern, ML in biophysics** (EPFL, 2018): recurrent autoencoders (Keras/TensorFlow) to decode polymer sequences from noisy nanopore current signals
- **Research Intern, ML for experiment automation** (MPI Stuttgart, 2017): neural networks to pre-screen scanning tunnelling spectroscopy data
- **Business Intelligence Analyst** (Royal Canadian Mounted Police, 2016): SQL and queueing-theory analysis of province-wide staffing data to benchmark detachment performance; initiated collaboration with UBC

SKILLS

- **Programming:** Python, Fortran (advanced); C++, Julia, bash (intermediate); SQL, R, MATLAB
- **Libraries and tools:** NumPy, JAX, PyTorch, pandas, Cython, MPI, OpenMP, CUDA, git, CI/CD
- **Quantitative methods:** Monte Carlo, variance reduction, stochastic optimisation, finite difference and finite element methods, numerical linear algebra, Bayesian methods, regularised regression, neural networks
- **Training:** HLRS Stuttgart courses in MPI/OpenMP, modern C++ design, CUDA and Julia HPC; Coursera *Financial Engineering and Risk Management* [in progress]
- **Languages:** English (native), German (B2), French (A1)

SELECTED PUBLICATIONS

Six peer-reviewed articles, three as first author, two more currently being drafted (see Google Scholar for up-to-date list), including:

- **Haupt** *et al.*, Optimizing Jastrow factors for the transcorrelated method, *J. Chem. Phys.* **158** (2023)
- **Haupt** *et al.*, Transcorrelated methods for multireference problems, *J. Chem. Phys.* **163** (2025)